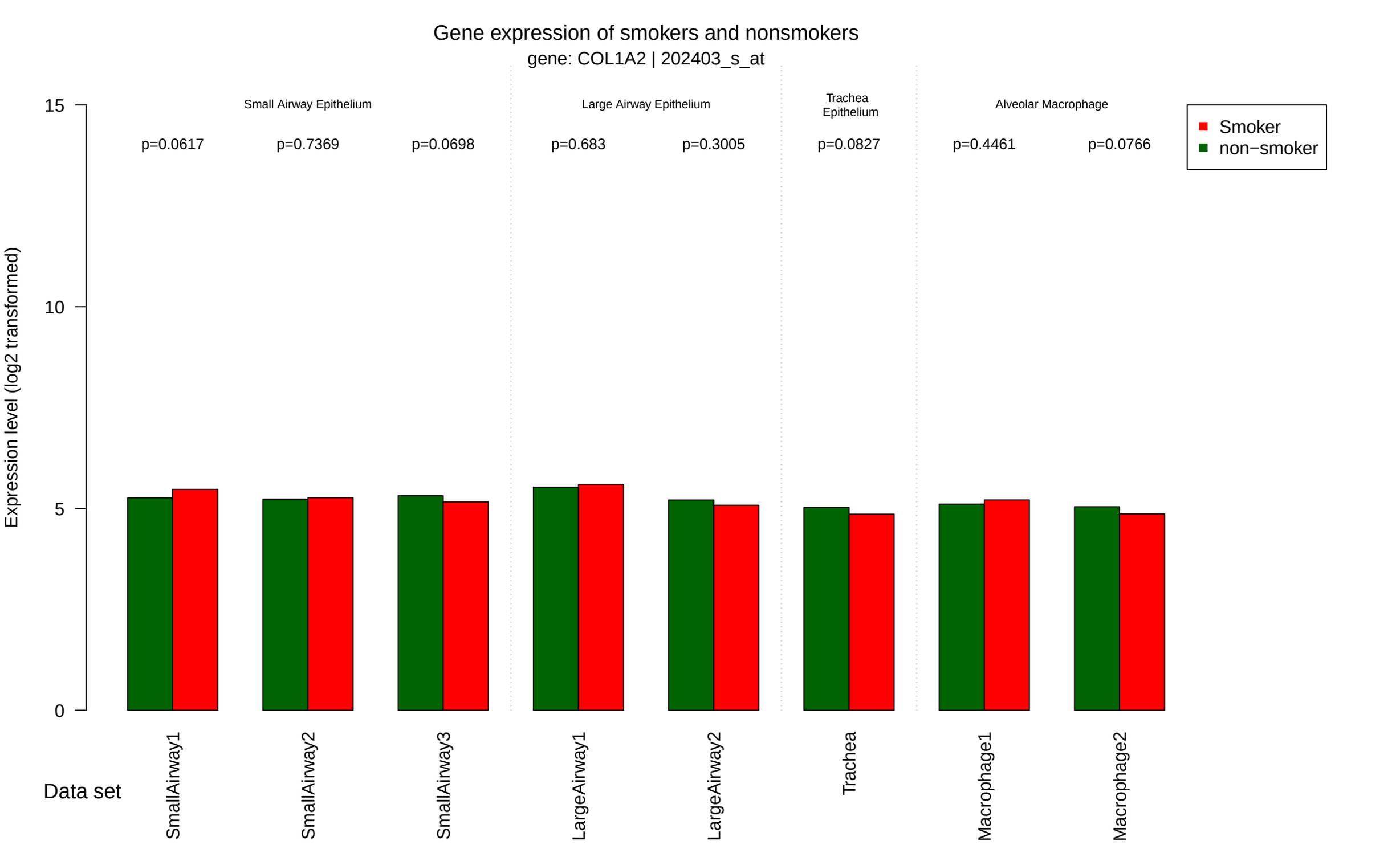
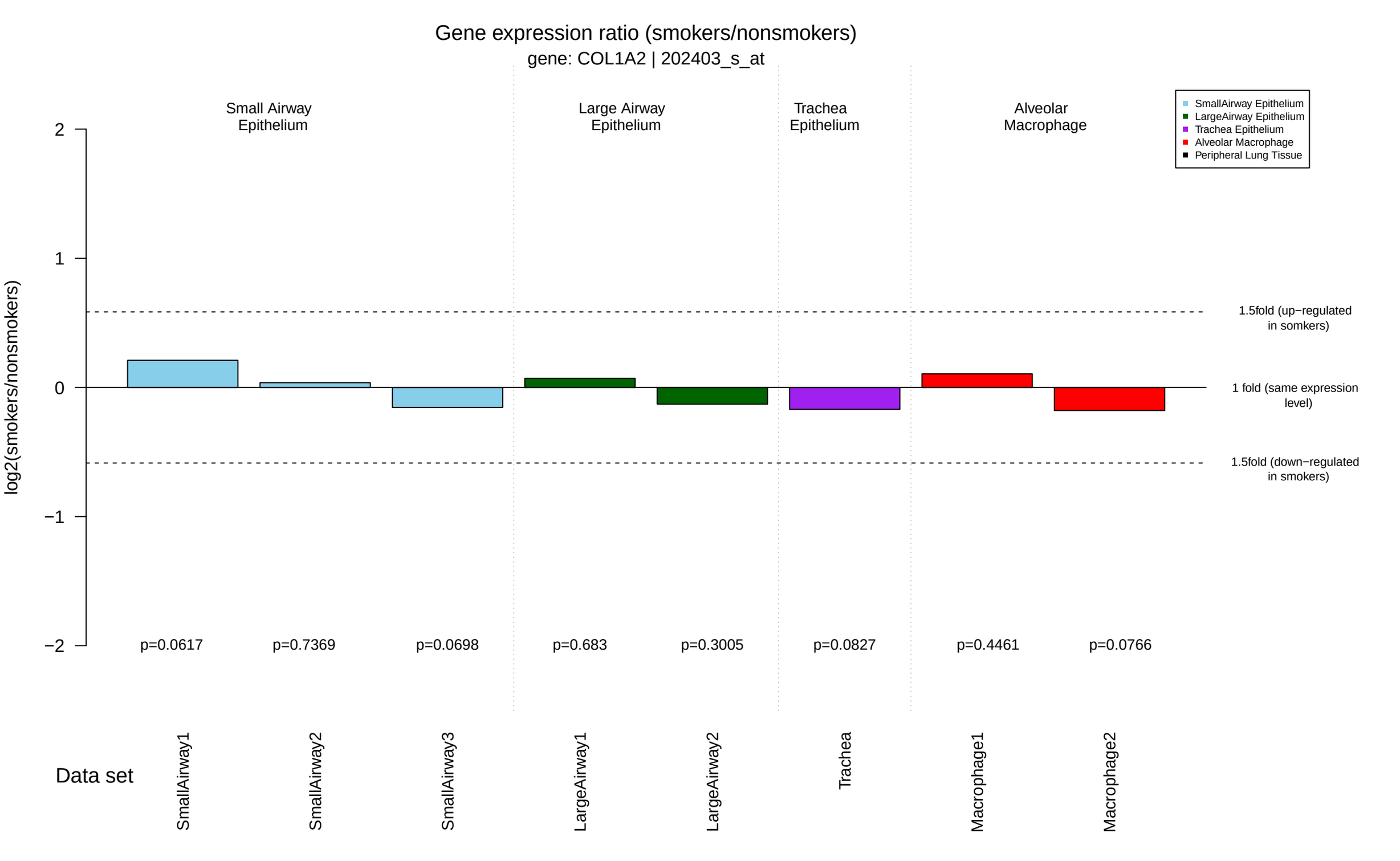
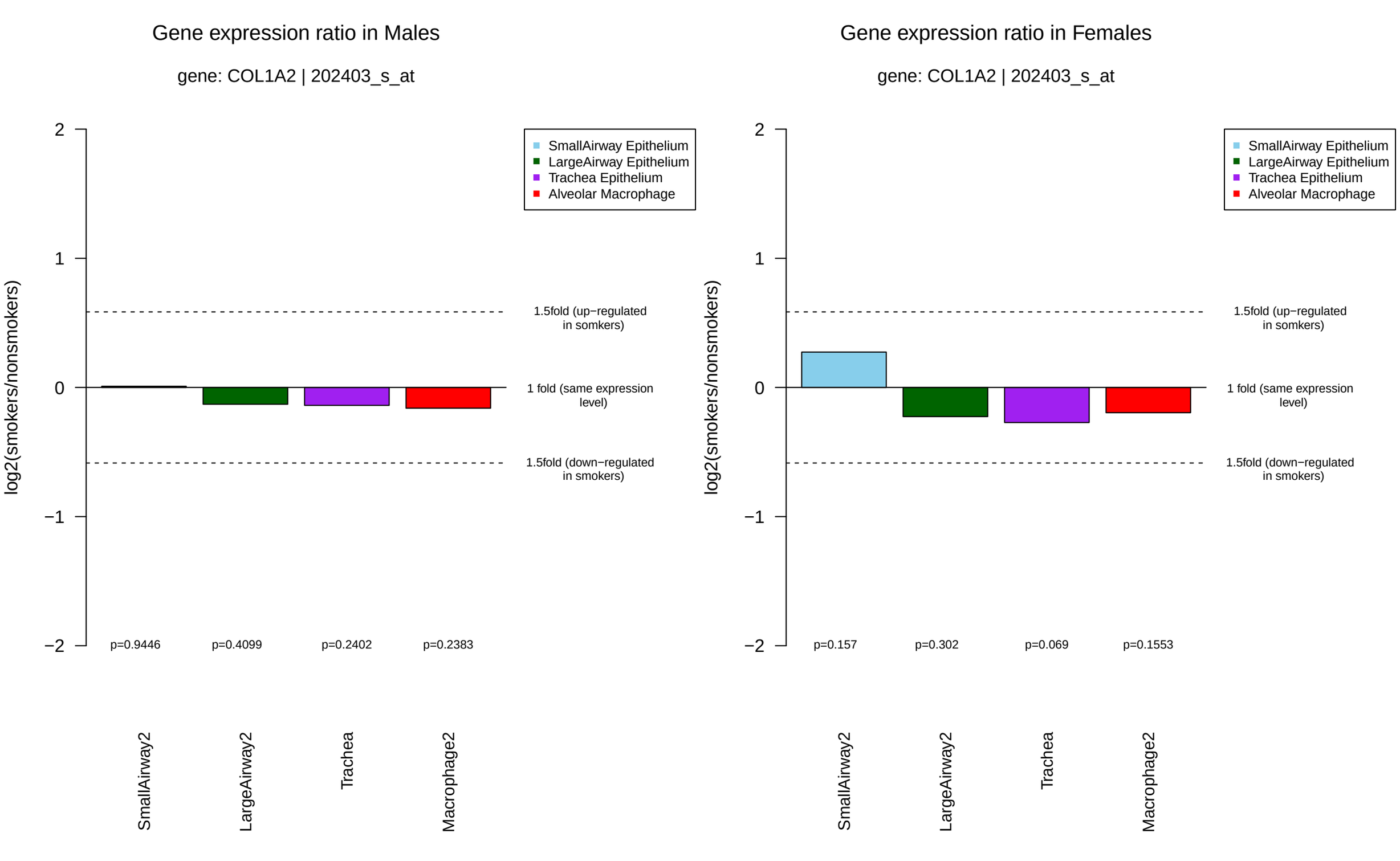
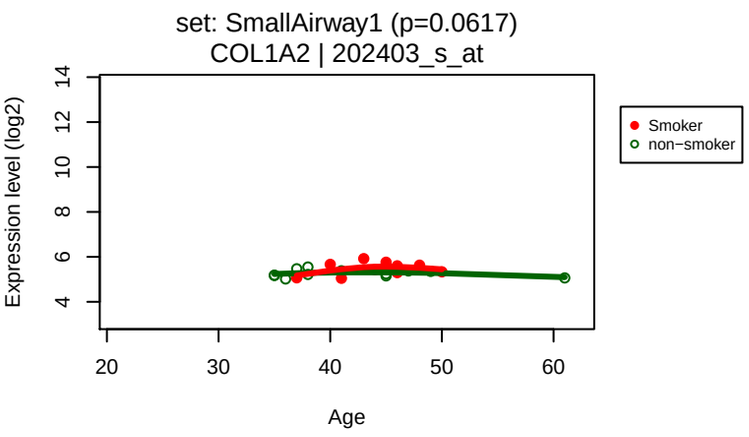
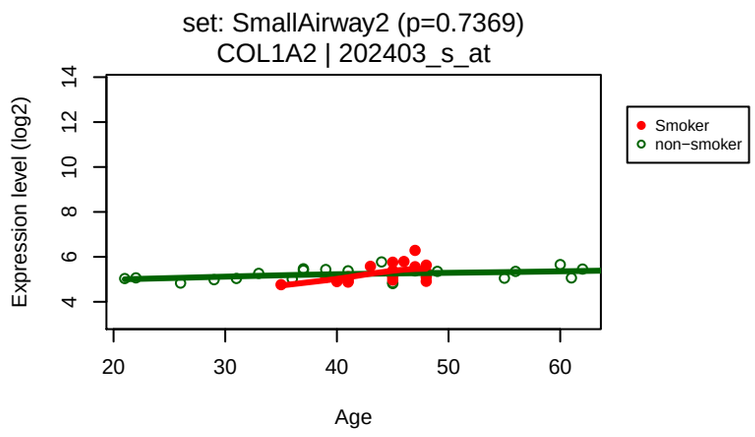
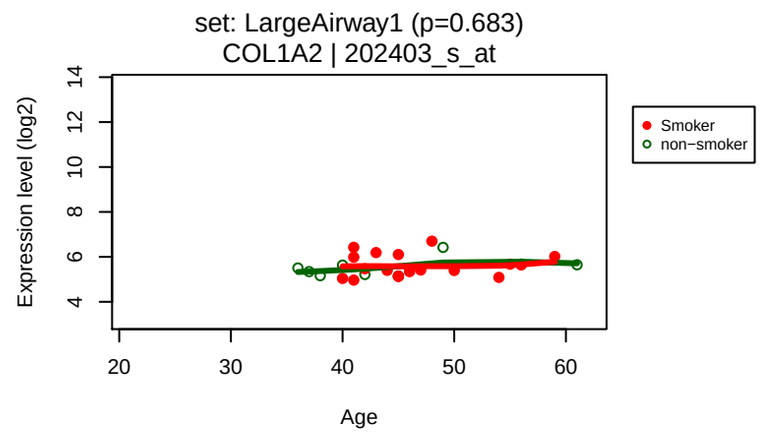

Gene expression of smokers and nonsmokers
gene: COL1A2 | 202403_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0617
p=0.7369
p=0.0698
p=0.683
p=0.3005
p=0.0827
p=0.4461
p=0.0766
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

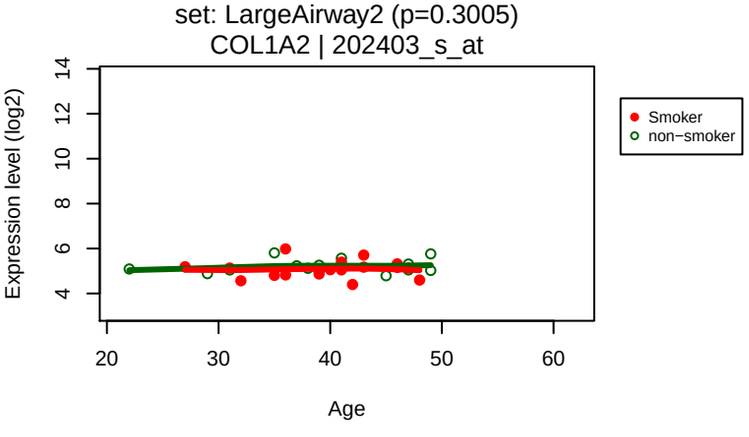

Gene expression ratio (smokers/nonsmokers)
gene: COL1A2 | 202403_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0617
p=0.7369
p=0.0698
p=0.683
p=0.3005
p=0.0827
p=0.4461
p=0.0766
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

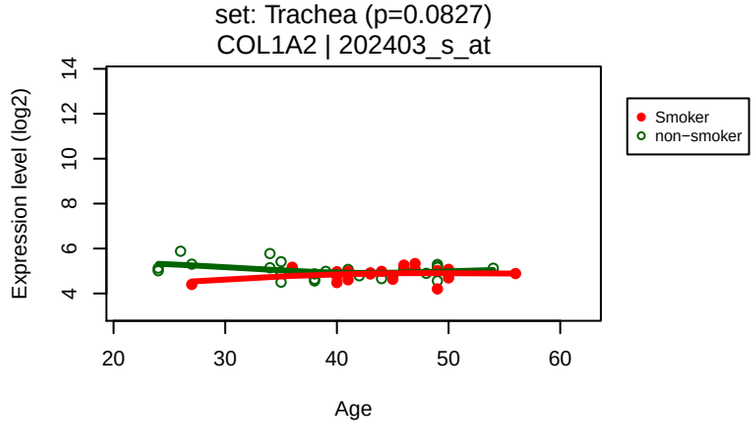

Gene expression ratio in Males
gene: COL1A2 | 202403_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9446
p=0.4099
p=0.2402
p=0.2383
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL1A2 | 202403_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.157
p=0.302
p=0.069
p=0.1553
SmallAirway2
LargeAirway2
Trachea
Macrophage2

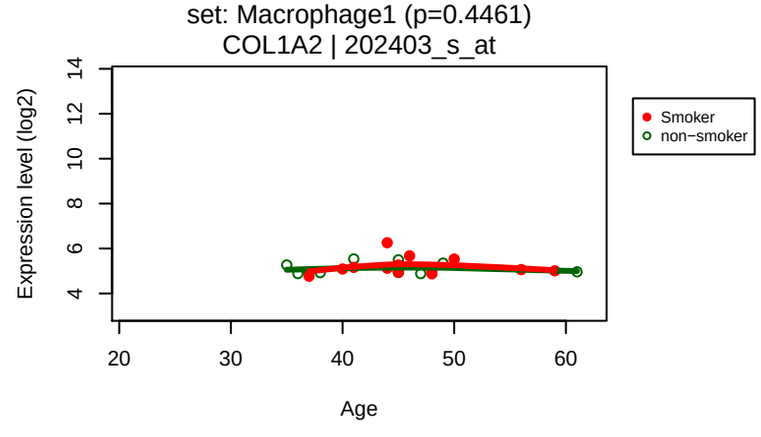

set: SmallAirway1 (p=0.0617)
COL1A2 | 202403_s_at
Expression level (log2)
Age
Smoker
non−smoker

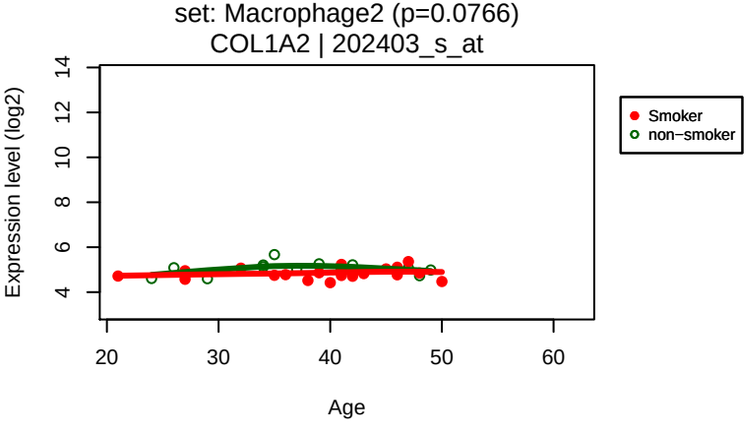

set: SmallAirway2 (p=0.7369)
COL1A2 | 202403_s_at
Expression level (log2)
Age
Smoker
non−smoker

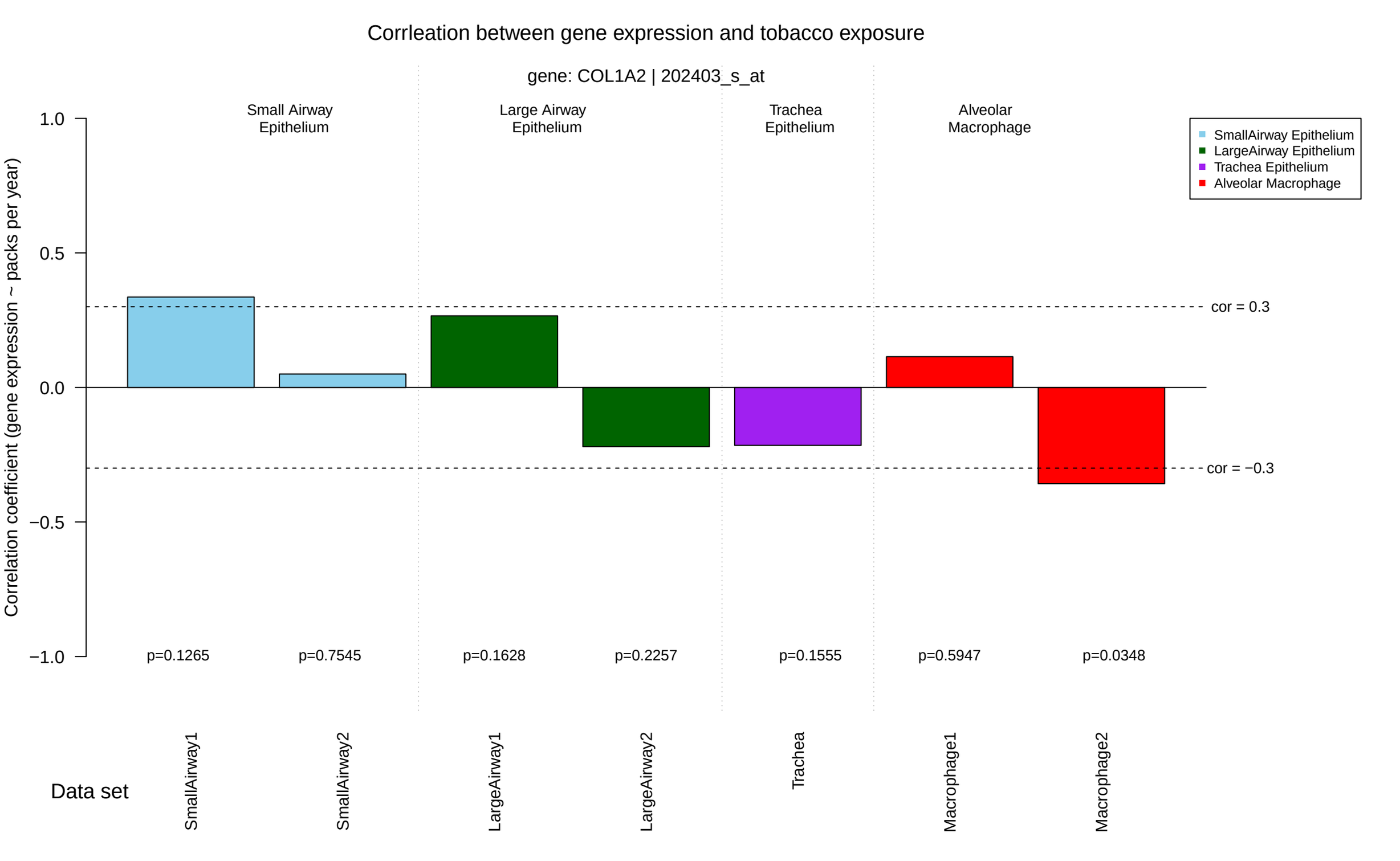

set: LargeAirway1 (p=0.683)
COL1A2 | 202403_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3005)
COL1A2 | 202403_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0827)
COL1A2 | 202403_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4461)
COL1A2 | 202403_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0766)
COL1A2 | 202403_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL1A2 | 202403_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1265
p=0.7545
p=0.1628
p=0.2257
p=0.1555
p=0.5947
p=0.0348
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2